Group (A) – Model Answers

1- Complete the following:

a. Inhaled air is rich in oxygen gas, while exhaled air is rich in carbon dioxide gas.

b. During breathing, exhaled air contains a higher amount of carbon dioxide.

c. The heart pumps blood carrying oxygen gas to all parts of the body.

d. In the alveoli, oxygen gas and carbon dioxide gas are exchanged.

e. Air enters through the nose or mouth and passes through the trachea until it reaches the lungs.

2- Answer the following:

a. What is the difference between inhalation and exhalation in terms of the amount of oxygen?

Inhalation brings oxygen-rich air into the lungs, while exhalation removes air containing less oxygen.

b. Describe the path of oxygen from the outside air until it reaches the body's cells.

Oxygen enters through the nose or mouth, passes through the trachea and bronchi into the lungs, reaches the alveoli, into the blood, and is carried by the heart through blood vessels to the body's cells.

3- Give reason for each of the following:

a. The chest moves up during inhalation.

Because the ribs move up and out and the diaphragm contracts and moves down, and lungs are filled with air.

b. The lungs contain a very large number of alveoli.

To provide a large surface area for efficient gas exchange.

4- Put (✓) for the correct statement and (✗) for the incorrect statement:

a. Oxygen leaves in the exhaled air, while carbon dioxide enters the body during inhalation. (✗)

b. The lungs expand during exhalation. (✗)

Group (B) – Model Answers

1- Complete the following:

a. The gas needed by the body to produce energy is oxygen.

b. Air enters the lungs during inhalation.

c. Blood carries oxygen gas to the whole body.

2- Answer the following:

a. Why does the respiratory system work as a team?

Because its organs work together to take in oxygen, exchange gases, and remove carbon dioxide from the body.

b. Explain how gas exchange takes place in the lungs.

In the alveoli, oxygen moves into the surrounding blood capillaries, while carbon dioxide moves from the blood into the alveoli to be exhaled.

3- Give a reason for each of the following:

a. Blood vessels surround the lungs.

To carry oxygen from the lungs to the body and transport carbon dioxide from the body to the lungs.

b. The chest moves up and down during breathing.

Because the ribs and diaphragm move to change the volume of the chest cavity, allowing air to enter and leave the lungs.

4- Put (✓) for the correct statement and (✗) for the incorrect statement:

a. Air enters the lungs during inhalation. (✓)

b. Air leaves the lungs during inhalation. (✗)

c. Inhaled air contains more oxygen than exhaled air. (✓)

Group (C) – Model Answers

1- Explain the process of breathing in humans. In your answer, mention:

• The organs of the respiratory system.

• What happens during inhalation.

• What happens during exhalation.

• How gases are exchanged in the alveoli.

Model answer:

- The respiratory system includes the nose and mouth, trachea, bronchi, and lungs.
- During inhalation, the diaphragm contracts and moves down, and the ribs move up and out, so air enters the lungs.
- During exhalation, the diaphragm relaxes and moves up, and the ribs move down and in, pushing air out.
- In the alveoli, oxygen moves into the blood while carbon dioxide moves from the blood into the alveoli to be exhaled.

2- Trace the path of oxygen during inhalation, naming the organs it passes through:

Model answer:

Nose or mouth → trachea → bronchi → bronchioles → alveoli, where oxygen passes into the blood.

3- Compare between:

Inhaled air and exhaled air

Model answer:

Inhaled air: contains more oxygen and less carbon dioxide.

Exhaled air: contains less oxygen and more carbon dioxide.

4- Choose the correct answer:

1. Which gas leaves the body during exhalation in large amount?

Correct answer: (B) Carbon dioxide.

2. Where does gas exchange take place?

Correct answer: (B) Alveoli.

3. After passing through the nose, air moves to the:

Correct answer: (B) Trachea.

4. The percentage of oxygen during exhalation:

Correct answer: (A) decreases.